AIR1 All-Zones Technical Overview

Smart i-LAB AIR1 All-Zones

Dashboard, shared API context, architecture, labels, pipeline, and verification

pctiope/smart-i-lab-testbed

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Objective

Review question

How does the AIR1 all-zones package turn two camera feeds and AIR-1 sensor rows into per-zone labels, a shared model, and an operator dashboard?

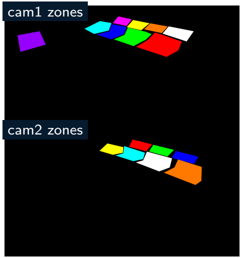
AIR1 only | All zones | No runtime secrets

- Keep the API/backend foundation minimal and scoped to AIR1 data access.
- Explain the two-camera, per-zone dashboard and label flow.
- Show the long-form `timestamp` + `zone_id` training contract.
- Keep Zone 5 CI/CD and deployment material out of this deck.

01

Dashboard And API Context

AIR1 All-Zones Dashboard



Tracked AIR1 zone masks define the dashboard's per-zone camera coverage.

Operational view

- One shared AIR1 model reports per-zone probabilities for zones 1–15.
- Two camera streams feed separate zone masks and one per-zone label table.
- The app can remain up while model readiness is blocked by a missing production pointer.

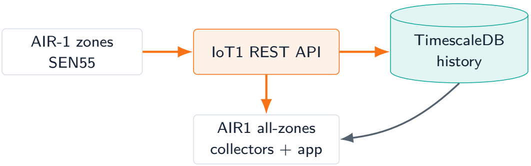
Important separation

- AIR1 excludes power and mmWave from model inputs.
- Missing camera coverage remains null; it is not converted to vacant.

Shared API Context

Purpose

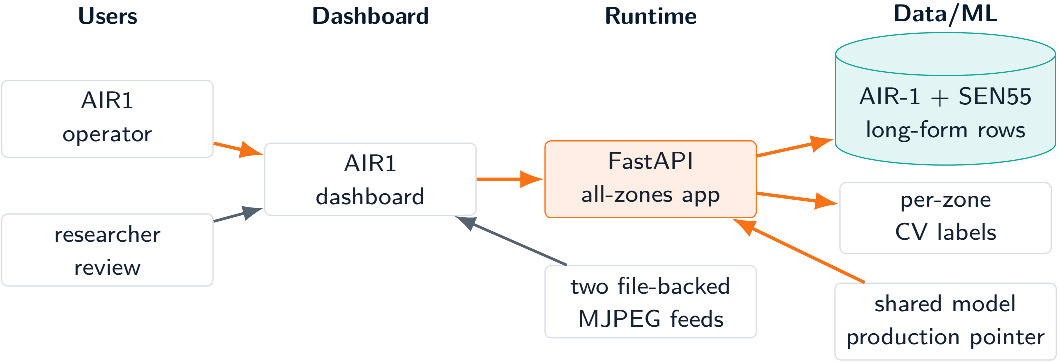
- AIR-1 and SEN55 data provide environmental feature rows for all model zones.
- Package-local collectors normalize source data into the all-zones contract.
- Runtime clients keep API keys and camera details in local environment files.



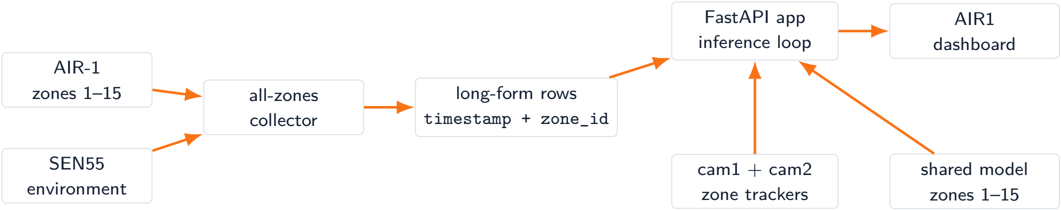
02

AIR1 Architecture

AIR1 User And Data Flow



AIR1 Runtime Architecture



Two cameras | Zones 1–15 | Long-form rows | No power/mmWave

03

Labels And Pipeline

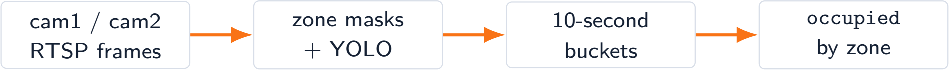
AIR1 Labels

CV truth

- Two cameras, per-zone masks, and one label table keyed by `timestamp` + `zone_id`.
- Target is `occupied`.
- Occupied when per-zone median `occupancy_count >= 1`.

Null handling

- A row with no matching CV label keeps `occupied = null`.
- Missing camera coverage is not converted into a negative label.
- Table 16 is excluded from the model-zone contract.



All-Zones Feature Contract

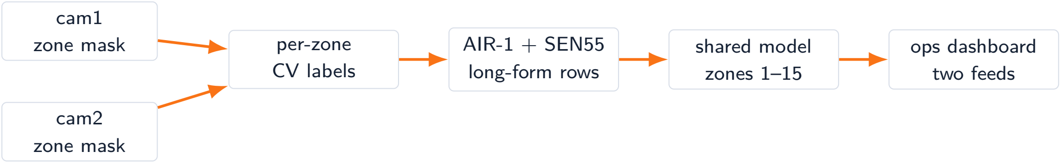
Input families

- AIR-1: temperature, humidity, CO2, PM2.5.
- SEN55: particulate, environment, VOC, NOx.
- Missing indicators remain part of the AIR1 feature contract.
- Time: hour and day-of-week cycles.

Contract guards

- Rows are long-form: one `timestamp` + `zone_id` pair per zone.
- Valid model zones are `1–15`.
- `zone_id` supports grouping, splits, audits, and display only.
- Power and mmWave are excluded.

AIR1 All-Zones Pipeline



Pipeline guard	Why it matters
Windows are built within each `zone_id`	Avoids mixing table histories inside one temporal sequence.
Production pointer gates inference	The UI can be healthy before a promoted model exists.
Retrain/promotion are explicit operations	Collection and dashboard uptime are not tied to automatic promotion.

04

Verification And Roadmap

Verification Gates

Contract checks

- `test_all_zones_contract.py` covers features, labels, zone masks, windows, and runtime payloads.
- Smoke checks reject candidates with the wrong target column.
- Package docs preserve the no-power/no-mmWave AIR1 boundary.

Runtime checks

- `/api/health` reports readiness, history size, and latest runtime errors.
- The dashboard can show camera freshness even when inference is waiting for a promoted model.
- Person counters keep writing annotated frames during upstream outages.

Roadmap And Risks

Near-term roadmap

- Promote a production AIR1 all-zones model after sufficient strict-window evidence.
- Keep tracker, collector, trainer, and app service health visible.
- Add model freshness and upstream API latency monitoring.

Risks to keep visible

- Label quality depends on two camera streams and per-zone mask coverage.
- Training quality depends on both-class labels across usable calendar windows.
- Missing production pointers should degrade inference, not dashboard availability.

Source Anchors

Topic	Primary source paths
Dashboard and runtime	`air1/.../web_app/`, `air1/.../run_live_app.sh`
Per-zone labels and masks	`air1/.../rtsp_zone_tracker.py`, `air1/.../masks/`
All-zones model contract	`air1/.../PIPELINE.md`, `air1/.../feature_contract.py`
Training and promotion	`air1/.../retrain_once.py`, `air1/.../promote_model.py`
Verification	`air1/.../tests/test_all_zones_contract.py`, `air1/.../smoke_test.py`

Closing point: AIR1 is an all-zones, two-camera contract with no Zone 5 power or mmWave signals imported into the model.